

IT1223(P) - Database Management Systems (P)
Department of Physical Science
University of Vavuniya
Lab Sheet - 06

i. How many Hotels are there?

Query:

```
SELECT COUNT(*) AS NumberOfHotels FROM Hotel;
```

```
MariaDB [hotel_db]> SELECT COUNT(*) AS NumberOfHotels FROM Hotel;
+-----+
| NumberOfHotels |
+-----+
|              5 |
+-----+
1 row in set (0.001 sec)
```

ii. What is the average price of a room?

Query:

```
SELECT AVG(Price) AS AverageRoomPrice FROM Room;
```

```
MariaDB [hotel_db]> SELECT AVG(Price) AS AverageRoomPrice FROM Room;
+-----+
| AverageRoomPrice |
+-----+
|          35.748333 |
+-----+
1 row in set (0.001 sec)
```

iii. What is the total revenue per night from all double rooms?

Query:

```
SELECT SUM(Price) AS TotalRevenue
```

```
FROM Room
```

```
WHERE Type = 'double';
```

```

MariaDB [hotel_db]> SELECT SUM(Price) AS TotalRevenue
  -> FROM Room
  -> WHERE Type = 'double';
+-----+
| TotalRevenue |
+-----+
|         160.00 |
+-----+
1 row in set (0.001 sec)

```

iv. How many different guests have made bookings for August?

Query:

```

SELECT COUNT(DISTINCT GuestNo) AS AugustGuests
FROM Booking
WHERE MONTH(DateFrom) = 8;

```

```

MariaDB [hotel_db]> SELECT COUNT(DISTINCT GuestNo) AS AugustGuests
  -> FROM Booking
  -> WHERE MONTH(DateFrom) = 8;
+-----+
| AugustGuests |
+-----+
|             0 |
+-----+
1 row in set (0.001 sec)

```

v. How many different guests have made bookings for May?

Query:

```

SELECT COUNT(DISTINCT GuestNo) AS MayGuests
FROM Booking
WHERE MONTH(DateFrom) = 5;

```

```

MariaDB [hotel_db]> SELECT COUNT(DISTINCT GuestNo) AS MayGuests
-> FROM Booking
-> WHERE MONTH(DateFrom) = 5;
+-----+
| MayGuests |
+-----+
|          5 |
+-----+
1 row in set (0.002 sec)

```

(d)

i. List the price and type of all rooms at the Grosvenor Hotel.

Query:

SELECT Price, Type

FROM Room

WHERE HotelNo = 'fb01';

```

MariaDB [hotel_db]> SELECT Price, Type
-> FROM Room
-> WHERE HotelNo = 'fb01';
+-----+-----+
| Price | Type  |
+-----+-----+
| 19.00 | single |
| 29.00 | double |
| 39.00 | family |
+-----+-----+
3 rows in set (0.001 sec)

```

ii. List all guests currently staying at the Grosvenor Hotel.

Query:

SELECT Guest.*

FROM Booking

JOIN Guest ON Booking.GuestNo = Guest.GuestNo

WHERE Booking.HotelNo = 'fb01' AND Booking.DateTo IS NULL;

```
MariaDB [hotel_db]> SELECT Guest.*
-> FROM Booking
-> JOIN Guest ON Booking.GuestNo = Guest.GuestNo
-> WHERE Booking.HotelNo = 'fb01' AND Booking.DateTo IS NULL;
```

GuestNo	GuestName	GuestAddress
10001	John Kay	56 High St, London

1 row in set (0.003 sec)

iii. List the details of all rooms at the Grosvenor Hotel, including the name of the guest

Query:

SELECT Room.*, Guest.GuestName

FROM Room

LEFT JOIN Booking ON Room.RoomNo = Booking.RoomNo AND Room.HotelNo = Booking.HotelNo AND Booking.DateTo IS NULL

LEFT JOIN Guest ON Booking.GuestNo = Guest.GuestNo

WHERE Room.HotelNo = 'fb01';

```
MariaDB [hotel_db]> SELECT Room.*, Guest.GuestName
-> FROM Room
-> LEFT JOIN Booking ON Room.RoomNo = Booking.RoomNo AND Room.HotelNo = Booking.HotelNo AND Booking.DateTo IS NULL
-> LEFT JOIN Guest ON Booking.GuestNo = Guest.GuestNo
-> WHERE Room.HotelNo = 'fb01';
```

RoomNo	HotelNo	Type	Price	GuestName
501	fb01	single	19.00	NULL
601	fb01	double	29.00	NULL
701	fb01	family	39.00	John Kay

3 rows in set (0.001 sec)

iv. staying in the room, if the room is occupied.

Query:

SELECT Room.RoomNo, Room.Type, Room.Price, Guest.GuestName

FROM Room

LEFT JOIN Booking ON Room.RoomNo = Booking.RoomNo AND Room.HotelNo = Booking.HotelNo

LEFT JOIN Guest ON Booking.GuestNo = Guest.GuestNo

WHERE Room.HotelNo = 'fb01';

```
MariaDB [hotel_db]> SELECT Room.RoomNo, Room.Type, Room.Price, Guest.GuestName
-> FROM Room
-> LEFT JOIN Booking ON Room.RoomNo = Booking.RoomNo AND Room.HotelNo = Booking.HotelNo
-> LEFT JOIN Guest ON Booking.GuestNo = Guest.GuestNo
-> WHERE Room.HotelNo = 'fb01';
```

RoomNo	Type	Price	GuestName
501	single	19.00	John Kay
501	single	19.00	Carol Farre
601	double	29.00	Mike Ritchie
601	double	29.00	Joe Keogh
701	family	39.00	John Kay

```
5 rows in set (0.001 sec)
```

v. What is the total income from bookings for the Grosvenor Hotel today?

Query:

SELECT SUM(Room.Price) AS TotalIncome

FROM Booking

JOIN Room ON Booking.RoomNo = Room.RoomNo AND Booking.HotelNo = Room.HotelNo

WHERE Booking.HotelNo = 'fb01' AND CURDATE() BETWEEN Booking.DateFrom AND
IFNULL(Booking.DateTo, CURDATE());

```
MariaDB [hotel_db]> SELECT SUM(Room.Price) AS TotalIncome
-> FROM Booking
-> JOIN Room ON Booking.RoomNo = Room.RoomNo AND Booking.HotelNo = Room.HotelNo
-> WHERE Booking.HotelNo = 'fb01' AND CURDATE() BETWEEN Booking.DateFrom AND IFNULL(Booking.DateTo, CURDATE());
```

TotalIncome
39.00

```
1 row in set (0.003 sec)
```

vi. What is the lost income from unoccupied rooms at the Grosvenor Hotel?

Query:

```
SELECT SUM(Price) AS LostIncome
```

```
FROM Room
```

```
WHERE HotelNo = 'fb01' AND RoomNo NOT IN (
```

```
    SELECT RoomNo FROM Booking WHERE HotelNo = 'fb01' AND CURDATE() BETWEEN DateFrom AND  
    IFNULL(DateTo, CURDATE())
```

```
);
```

```
MariaDB [hotel_db]> SELECT SUM(Price) AS LostIncome  
-> FROM Room  
-> WHERE HotelNo = 'fb01' AND RoomNo NOT IN (  
->     SELECT RoomNo FROM Booking WHERE HotelNo = 'fb01' AND CURDATE() BETWEEN DateFrom AND IF  
NULL(DateTo, CURDATE())  
-> );  
+-----+  
| LostIncome |  
+-----+  
|      48.00 |  
+-----+  
1 row in set (0.002 sec)
```